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Modeling virtual knowledge graphs using relevant news data by NLP methods for business analysis

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***Abstract*—Companies aiming at online data sources for extracting business insights often experience the issue of managing different collections of datasets. It is because online business-related data is growing over time. This increasing data demands more storage. But saving the irrelevant data will cost more to companies. To overcome the problems of data integration and scalability, the idea of Virtual Knowledge Graphs (VKGs) is exploited. Moreover, text processing methods based on Natural Language Processing (NLP) are utilized for improving the identification of relevant business news data. The contribution of this work is twofold that is presented using a business case study. First, we have applied the concept of VKG to expand the company database. The expansion is required for adding more business-related data from the news articles to improve the process of identifying its relevance as per the end-user's interests. Second, we have proposed the text relevance resolution matrix and its working is shown through NLP techniques for identifying relevant news data. Populating the modeled VKG with relevant news data will save database space and prevent manual labor. Moreover, it will also help business analysts to understand business data temporally, spatially, and thematically.**

***Index Terms*—Artificial Intelligence (AI), Business Intelligence (BI), Natural Language Processing (NLP), Text Relevance**

I. Introduction

The majority of companies experience the issue of managing huge collections of datasets [1]. Often, such companies have different departments which work independently and produce data silos, which are not interlinked but include semantically-related data. This data probably has redundancy and inconsistency [1], [2]. To efficiently utilize this data, it is important to integrate the data silos. Integrating data silos, scaling them based on current demands, and providing appropriate access to departments are labor-intensive and expensive [1]. Hence, many business analysts integrate data in an ad-hoc manner, which is an inefficient, time-consuming, often results in data quality problems, and negatively impacts business decision-making. To overcome the problems of data integration and scalability which arise using traditional solutions based on the relational model, the idea of VKG [1] is used in this research. The VKG substitutes the rigid structure of data tables by providing us with the flexibility of graphs that are kept virtual [1] and contain embedded business knowledge. To show the proof-of-concept modeling of a VKG, an industrial case study was considered. This case study involves a company that is working on news articles to identify present business opportunities. Their objectives are; 1) to identify the relevant business data and classify it into pre-defined themes based on a company taxonomy (i.e. a hierarchy of organized business keywords), 2) to store the relevant news data in their database, and 3) to report the business opportunities to end-users based on their interests. Currently, the identification of the relevant news data and its classifications are executed manually. In addition, presently, the only method of detecting the relevant news data is through a business taxonomy that was created many years before. The contribution of this article based on this case study is twofold. First, is the presentation of a VKG to expand their current database. Second, the proposal of the text relevance resolution matrix for identifying relevant news data to populate the modeled VKG. The rest of the paper is organized as follows: In Section 2, the background is discussed. In Section 3, a business case study is presented. Section 4 discusses the modeling of VKG and the proposal of the text relevance resolution matrix. In Section 5, the presented system is discussed. Finally, in Section 6, the conclusion of the article is presented.

II. Background

A VKG system offers access to a relational database through an ontology [2]. Here, ontology presents users with high-level data views which are not technical but constructed using vocabulary which is familiar to them [2]. The language to express an ontology is the W3C Web Ontology Language (OWL2) [1], which allows us to model a hierarchy of classes. The ontology is connected to data sources using mapping [2]. This mapping includes a group of database queries that can be used for querying a dataset. Also, it specifies how the classes of ontology are populated by data values from the source database. In addition, the data in the VKG is organized in the form of a Graph (G) where the nodes are the representation of domain objects and data values and the edges of a graph represent the properties of objects. Moreover, the VKG offers the enrichment of semantic information by defining concepts and property hierarchies, and a domain

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and range of properties [1]. A unique characteristic of VKG systems is that they extract data from relational data sources only when it is needed for a specific user query, rather than retrieving all the data. This means that the knowledge graph remains virtual [2]. In the existing literature, the VKGs have been exploited for many applications across different domains. However, some of them are selected as follows; Lu et al [3] proposed a framework to automatically construct VKGs using unstructured documents. Kurniawan et al [4] proposed dynamic construction of VKGs for federated log analysis. In addition, Calvanese et al. [5] proposed the onprom framework based on VKGs using the extraction of event logs from legacy information systems. Their purpose in building VKGs was to improve process mining-based tasks for business applications. Moreover, Güzel et al. [6] presented Ontop-temporal based on VKGs for query answering with temporal data and ontologies. Furthermore, Bruggemann et al. [7] used VKGs for combining and processing real-time and static data from different sources for the maritime domain. The usefulness of VKGs is evident from the above-mentioned applications. Where, based on different domains, the VKGs were populated with data. But, the question that arises here is, what is the criteria that should be set for selecting the data for populating the VKGs? For selecting the data, most existing systems [8] involve binary data classification, i.e., texts classified as relevant or irrelevant. In the literature, text relevance refers to the match between an analyst's goal and textual information useful to that goal [9]. Information that matches an analyst's goal is more relevant. Whereas, information that does not match the goal is less relevant, irrespective of its importance. Most commonly, there are two methods [8], [9] to find relevant texts, which are; 1) keyword-based approach and 2) machine learning-based classifiers [8], [9]. In keyword-based classification, the news articles are categorized and identified as relevant using a list of business-related keywords. There exist other systems that rely on machine learning classifiers, such as Bayesian algorithms and Support Vector Machines (SVMs). Existing literature also covers classification systems that are built using keyword matching techniques combined with machine learning methods. Despite several state-of-the-art techniques to find relevant data, interestingly, there is no formal definition exists [8], [9] that attempts to quantify the level of text relevance. To fill this research gap, the text relevance resolution matrix is proposed.

III. Case Study

The case study involved the French company FirstECO, whose objective is to identify existing business opportunities in online news articles, classify them based on pre-defined themes, and finally send them to end-users according to their interests. The firm publishing its intention in the news article of acquiring the vacant land for its expansion in the region is a potential business opportunity for individuals or other firms who want to sell or rent their land. For identifying business opportunities, at present, the company is scraping the data of news articles from the official news websites and storing them in their relational database. To categorize business opportunities into different domains, the experts in the company perform text classification of the news articles. For classifying texts, they rely on the manual method of identifying themes of the news articles and storing them along with the articles in a database. The manual text classification process involves an expert who understands the textual content of the news article and classifies it accordingly. This method gives good results but it is expensive and time-consuming. To improve the quality of business data analysis, the company wants to; 1) expand their existing relational data (i.e. data of news articles) without moving or transforming but integrating it with the associated data (i.e. semantic information) of the entities (people, company details, time, and location) mentioned in the news articles to improve news data relevance identification, and 2) automate a method of identifying the relevant news data using text processing methods data to populate their database.

IV. Proposed Method

The proposed method deals with; 1) the modeling of a VKG, and 2) populating the modeled VKG with the relevant news data defined by the text relevance resolution matrix.

A. Modeling of a VKG

A VKG requirement is a tuple P = (O, M, S), where O is an ontology, S is a schema of a data source and M is a mapping from a data source schema (S) to ontology (O) [1], [2]. To model a VKG, first, we need a schema of a company database. For protecting the company data's confidentiality, we used only three data tables which are; Article, Source, Company from the company database (see "Fig. 1"). These tables are linked using primary and foreign keys.

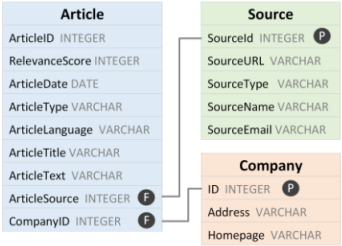

Fig. 1. Database schema of a company.

Once we have the database schema, the concepts are selected (see "Fig. 1") for designing the business ontology for constructing an RDF graph. Business ontology provides us with a formal description of knowledge using concepts within a domain and the relationships that are held between them. Based on the existing company database schema, the properties of the concepts of ontology are defined. Moreover, to enrich the company's database with additional business-related data that can be extracted from the news articles, new properties such as address, number of employees, email, homepage, type of industry, the last turnover, products, services, location, homepage, domain, a person of interest, writer, and the publisher are added after the literature review. In this way, we are trying to transform as well as extend the company

database. For this, a collection of classes and properties from state-of-the-art ontologies and controlled vocabularies (i.e. organized arrangement of words describing specific domains) are selected (see 'Fig. 2") for defining the concepts and their properties, which are; 1) Resource Description Framework Schema (RDFS), DCTERMS, DCAT ontology, and vCard ontology. In addition, the model uses GeoNames for specifying the spatial features.

After defining the concepts and the properties of the business ontology, we will transform the relational tables into RDF graphs through mappings. An RDF graph is a set of triples that is defined using nodes and edges. An RDF triple has three components which are; subject, predicate and object. For a database mapping, in the relational table, each table represents one entity type, and the rows of the table represents the instances of the entity type for an RDF graph. For demonstration, a small portion of an RDF graph is shown in "Fig. 3". In this way, the relational tables are linked in an RDF graph.

In the existing literature [1], [2], R2RML i.e. RDB to RDF Mapping Language is used to create the mappings between the relational data tables and RDF graphs for constructing VKGs. The R2RML consists of a triples map and a subject map. A triples map is a mapping from one row of a table to RDF triples. Whereas, a subject map defines the subject that is shared by all triples output from that triples map. Moreover, triples are generated based on predicate-object maps. "Fig. 4" provides a high-level view of using the R2RML for our case.

B. Identifying relevant news data for a VKG

To identify the relevant news data, we need to define the term "relevance" in the context of business data. For this, the text relevance resolution matrix is constructed (see "Fig. 5") to define four different levels of text relevance that can be used for filtering the news data of interest by the company. The 1st level of identifying relevant business data is maintaining a vocabulary of known domain words (i.e. predefined taxonomies) which are specific to business entities describing the interests of the company.nThe 2nd level of text relevance addresses the limitation of taxonomy by incorporating the semantic relationships among words for extracting the topics of the news articles. The extracted topics (i.e. a collection of keywords) characterizing the news articles are later mapped with the company taxonomy or a dictionary for identifying relevant texts. The 3rd level of text relevance is to identify the news articles using named entities, e.g. person names, organizations, etc. This level of text relevance will also include the identification of domain-specific entities (i.e. thematic entities describing an overall document's context as topics). Lastly, the 4th level of text relevance deals with the identification of business events (i.e. a combination of thematic entities, spatial and temporal entities along with the information of the named entities). These events are stored as knowledge graphs which are constructed using business ontologies.

In context of business analysis, it is also important to identify relevant data based on different levels of geographical entities such as countries, regions, cities, etc. Using spatial resolution, the criteria can be set to filter the news articles based on the geographical location, i.e. country, region, department, or city. In addition, by including the temporal resolution, the dates and the timestamps can be found in the news articles for clarifying them based on the required temporal expressions (e.g., 17 Aug. 2022). Lastly, to include the semantic content of the news articles' texts for finding their relevance to the company, four levels of semantic information are defined. The 1st level is to include word embeddings (i.e. a representation that allows words having a similar meaning to have a similar representation) of keywords defined in the company's taxonomy or a dictionary. The 2nd level is to take in sentence embeddings that represent entire sentences and their semantic content as vectors. This information helps systems to understand the context of the entire text. The 3rd level of semantic information is to consider document embeddings, and the 4th level includes embeddings of the graphs constructed using named entities. These knowledge graphs contain business events.

C. Working of a Text Relevance Resolution Matrix

To show the working of the text relevance resolution matrix for identifying different levels of relevant business data, we need the data of business entities or articles which are classified as relevant by the company. Using this data as the gold standard, we can show different levels of text relevance of new business articles. For this, the company has provided us with two datasets, which are; 1) The business taxonomy of around 3,000 business keywords. The keywords are classified into two categories, which are "topics" and "sectors". 2) The dataset covers 20,000 text files of the relevant news articles which were manually identified by the company over the last few years. This dataset was collected from the 1) regional websites (dedicated websites for French regions, 2) news websites, and 3) business companies websites. In the next section, it is shown how the two company datasets are exploited for detecting different levels of the relevant news articles using NLP methods.

1st level of text relevance: The 1st level of text relevance deals with the identification of the keywords defined in the taxonomy in the news articles. To automate this process of recognizing relevant business data, the application of state-of-the-art machine learning-based models for text processing is explored. The data received from the company contains only the collection of relevant business keywords that is organized as a taxonomy. The other class of irrelevant keywords is missing in this case. Based on the data availability, the process of detecting the relevant news articles is treated as a one-class classification problem. For this, we used a one-class Support Vector Machine (SVM) for a novelty detection algorithm designed by the Scikit-learn team [8]. To show the work of a one-class SVM for detecting whether a business document is relevant or not, a set of keywords belonging

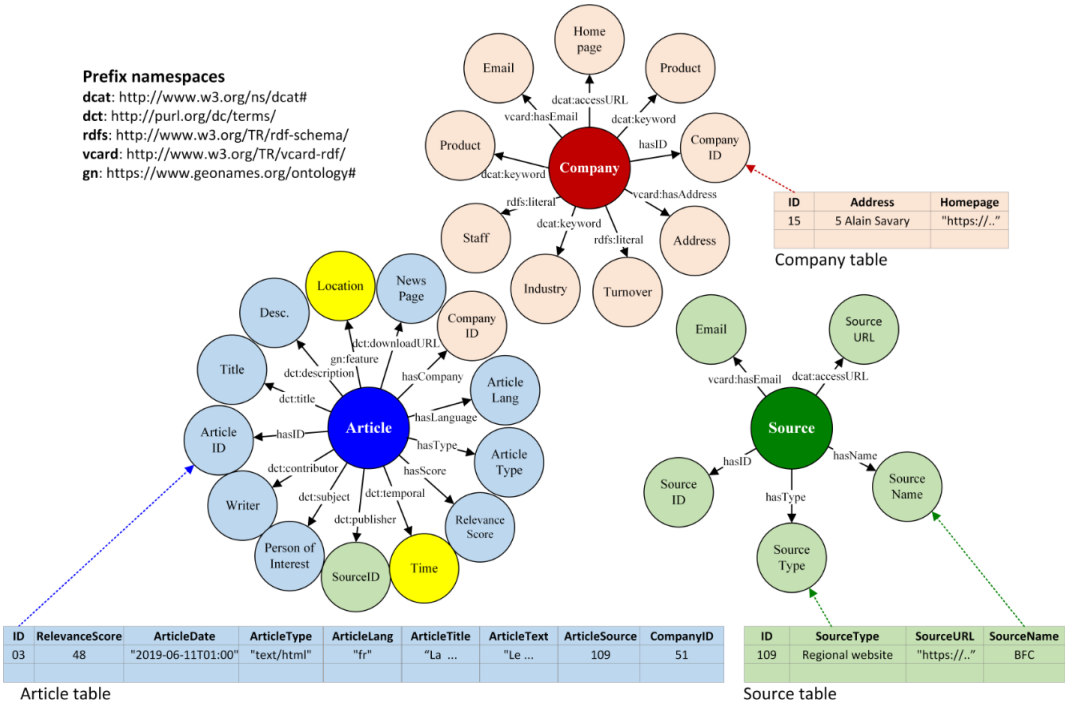

Fig. 2. Modeling a VKG.

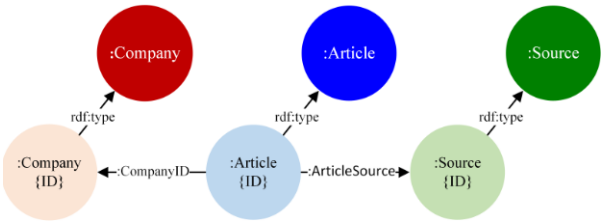

Fig. 3. An RDF graph based on relational data.

```
@prefix dct:<http://purl.org/dc/terms/> .
@prefix rr: <http://www.w3.org/ns/r2rml#> .
@prefix ex: <http://example.com/> .
@base <http://example.com/base/> .

:ArticleMapping a rr:TriplesMap ;
    rr:logicalTable[ rr:sqlQuery "SELECT * FROM Article" ] ;

    rr:subjectMap  [ rr:templete "http://example.com/Article{id}" ;
                     rr:class :Article
    ] ;
    rr:predicateObjectMap [ rr:predicate     :hasID ;
                            rr:objectMap [ rr:column "ID" ]
    ] ;
    rr:predicateObjectMap [ rr.predicate  dct:title ;
                            rr:objectMap [ rr:column "ArticleTitle" ]
    ] .
```

Fig. 4. R2RML.

to two different categories (topics and sectors in our case) of the company taxonomy are considered. To feed the one-class SVM model, the keywords of taxonomy are transformed into vector representations. Using the vector representations of these keywords, the model is trained and later the keywords, which are the features extracted using the Count Vectorizer (i.e. a process of transforming a given text into a vector based on the word frequency) from the new articles, were

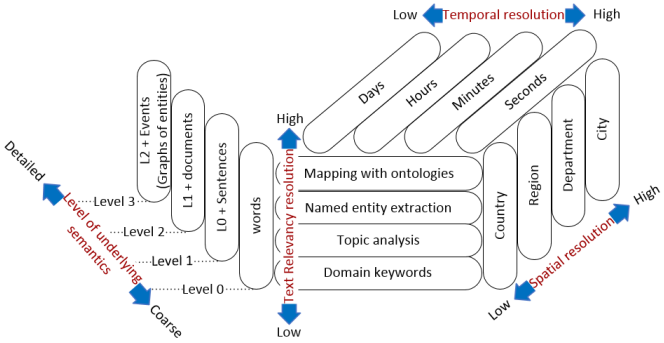

Fig. 5. Text Relevance Resolution Matrix.

investigated to check their relevancy against the two different categories of taxonomy concepts. Here, a score can be set based on the number of keywords belonging to the learned keywords boundary region (see "Fig. 6"). In the case of a higher number of irrelevant keywords, the news articles can be discarded.

Once the news document is treated as a relevant document, we can find out its relevancy against the certain taxonomy concepts which are similar to them. This will help us to find out the subcategory (trading, construction, rental, etc.) of a news article as per the business taxonomy. To calculate its relevance in percentage, the word vectors of taxonomy concepts are compared with the word vectors (features identified using the Count Vectorizer by Scikit-learn) of a news article. Using the Cosine similarity i.e. measured by the cosine of the angle between two vectors and determines whether the two vectors

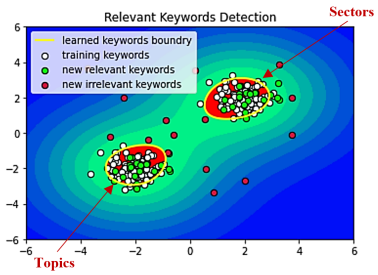

Fig. 6. Relevant keywords detection using one-class classifier.

are pointing in the same direction. The similarity scores among the word vectors are computed and later the mean of these scores is calculated to obtain an overall similarity percentage value (see "Fig. 7").

Taxonomy ID
Subcategory
Category

	ConceptFR	ConceptEN	ConceptID	Type	ValueSimilarity
0	Fermeture de site	site closure	2187	Topics	0.2400
1	Préparation de site	site preparation	23923	Sectors	0.2400
2	Service immobilier	real estate service	5698	Sectors	0.2300
3	Regroupement de plusieurs sites	consolidation of several sites	4720	Topics	0.1900
4	Vente d'espace publicitaire	sale of advertising space	10666	Sectors	0.1800

Fig. 7. TF-IDF-based news data relevance detection.

2nd level of text relevance: The 2nd level of text relevance addresses the limitation of the 1st level of text relevance by incorporating the semantic relationships among words for extracting the topics (i.e. cluster of words) of the news articles. This made possible using state-of-the-art BERTopic model [11]. It is a topic model that use a class-based variation of TF-IDF (i.e. a measure to represent the significance of a word to a document). It creates document embedding with pre-trained language models, clusters these embeddings, and produces topic representations. To use BERTopic model [11] for identifying the relevant news articles, the 1st step is to preprocess the dataset of relevant 20,000 news articles provided by the company. Then, sentence-transformers are used for capturing document-level embeddings by transforming the text files into 512-dimensional vectors. For acquiring the topics, the BERTopic model is initialized, trained on the corpus of relevant news articles and topics with c-TF-IDF scores are extracted as clusters (see "Fig. 8"). After training the BERTopic model, the topics of new documents can be predicted. The predicted topics are compared with the keywords of the business taxonomy using the Cosine similarity to understand its relevancy to the company.

3rd level of text relevance: Finding topics from text in the 2nd level of text relevance helps to find the theme of a news article. The themes will help to filter the relevant news articles based on the company interest. However, along with themes of news articles, companies also need to know the main entities

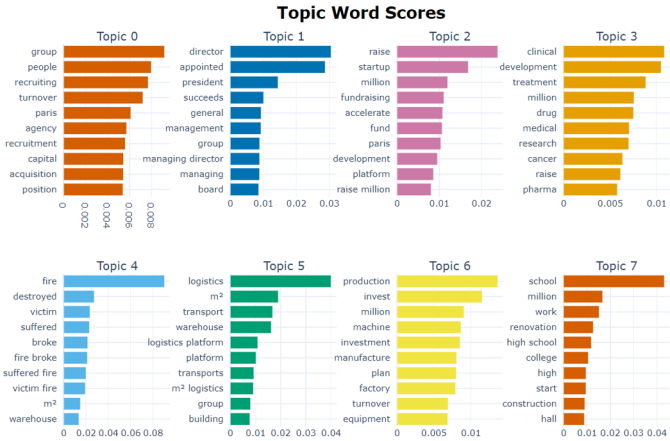

Fig. 8. BERTopic-based news data relevance detection.

(persons names, organizations, locations, etc.) mentioned in the news articles. The entities will help companies to understand the origin of the news information. Named-entity recognition (NER) i.e. a subtask of information extraction is used for finding and classifying named entities mentioned in text into pre-defined categories such as person names, organizations, locations, time expressions, etc. There are many Java and Python-based libraries available for performing NER on text. A few of them are SpaCy, Apache OpenNLP, and TensorFlow. SpaCy [12] is selected here to show the proof-of-concept named-entity extraction. Using SpaCy's NER library [12], the named-entities (location, time and person name) are identified in the "Fig. 9". The identified named entities will help to filter the relevant news articles based on the time, location and interested persons of interest as required.

Fig. 9. NER-based news data relevance detection.

4th level of text relevance: The 4th level of text relevance comes into action when articles are filtered into relevant and irrelevant categories using the combination of thematic and spatio-temporal entities of news articles. These entities are grouped to form business events (i.e. a thing that happens at a certain time and a certain location by a certain company). To understand business events, we need the fundamental information of the company (i.e. title, address, number of employees, email, homepage, type of industry, the latest turnover, products, and services) and the content of the news article (i.e. the title, description, location, homepage, domain, a person of interest, writer, and the publisher). For constructing business events, first, we need to extract the information about the above-mentioned entities using NER as we did in the 3rd level of text relevance. Second, to store the extracted

entities as knowledge graphs to represent business events, we need to define the relations among them. The modelled VKG enables us to store the extracted information using state-of-the-art ontologies and controlled vocabulary. After storing the data in the VKG, the queries can be written for extracting business events. For instance, "Fig. 10" shows the combining of spatial and temporal entities along with the thematic entities for studying the evolution of different business events over the years.

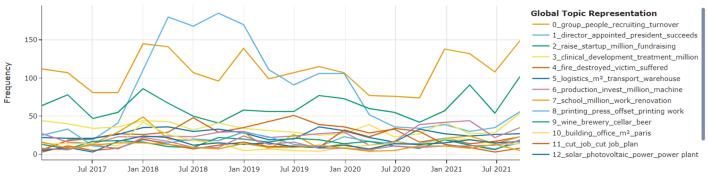

Fig. 10. Combining thematic and temporal entities to understand the evolution of business data.

V. Discussion

The article presents the modeling of a VKG with spatial, temporal, and thematic entities that can be extracted from the news articles through NLP. Initially, the fundamental requirements to understand the business opportunities were identified. These requirements were mainly based on 3 types of information, which are; 1) news article information, 2) company information, and source information. Moreover, information concerning the entities related to the news articles, the company, and the source is also considered. The reason for adding company-related entities is because of their significance for evaluating business opportunities by considering the company's resources, the type of resources, and its position in the business market. Based on these identified entities, a VKG is modelled using state-of-the-art ontologies and controlled vocabularies by extending the current database of a company.

The second step is to identify the relevant news articles to populate the modelled VKG. For this, we defined the text relevance resolution matrix. Using this matrix, four different levels of text relevance can be used for filtering the news articles of interest by the company. To supplement the text relevance, the spatial and temporal entities are incorporated into the text relevance matrix. The spatial entities will reveal whether a business event is local or global. Also, the temporal entity will indicate the business news popularity and duration. Eventually, the linking of the spatial, temporal, and thematic entities of business events will reveal valuable business insights for the end users. For instance, if we combine the thematic and temporal entities of news articles, we can observe the evolution of market trends over the last few years (see "Fig. 10"). As it can be seen dominant business opportunities during the middle of 2021 were related to "recruiting people" and "startups" (see "Fig. 10").

VI. Conclusion

In this paper, we have shown how VKGs can be used for providing a unified view of company data by overcoming the problems concerning data integration and expansion. In addition, the text relevance resolution matrix is proposed using NLP techniques for identifying relevant news data for populating KVG. The idea behind the text relevance resolution matrix is to filter the news data using different spatial, temporal, and thematic matrices. For example, it is crucial to know for how long a certain business opportunity has been active. How is it evolving locally, or worldwide? Or how did different companies come up with this idea as competitors in the market? The presented system aimed at addressing these questions by supplementing the VKG with relevant news-related data. Moreover, the proposed text relevance resolution matrix can serve as the foundation for future data classification problems. Such problems are centered on filtering the text documents into relevant and irrelevant categories not only for business analysis but also applicable in other domains. The limitation of this ongoing work is the validation that is currently missing in this article. Future work can be done to evaluate the system efficiency for industrial case studies.

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